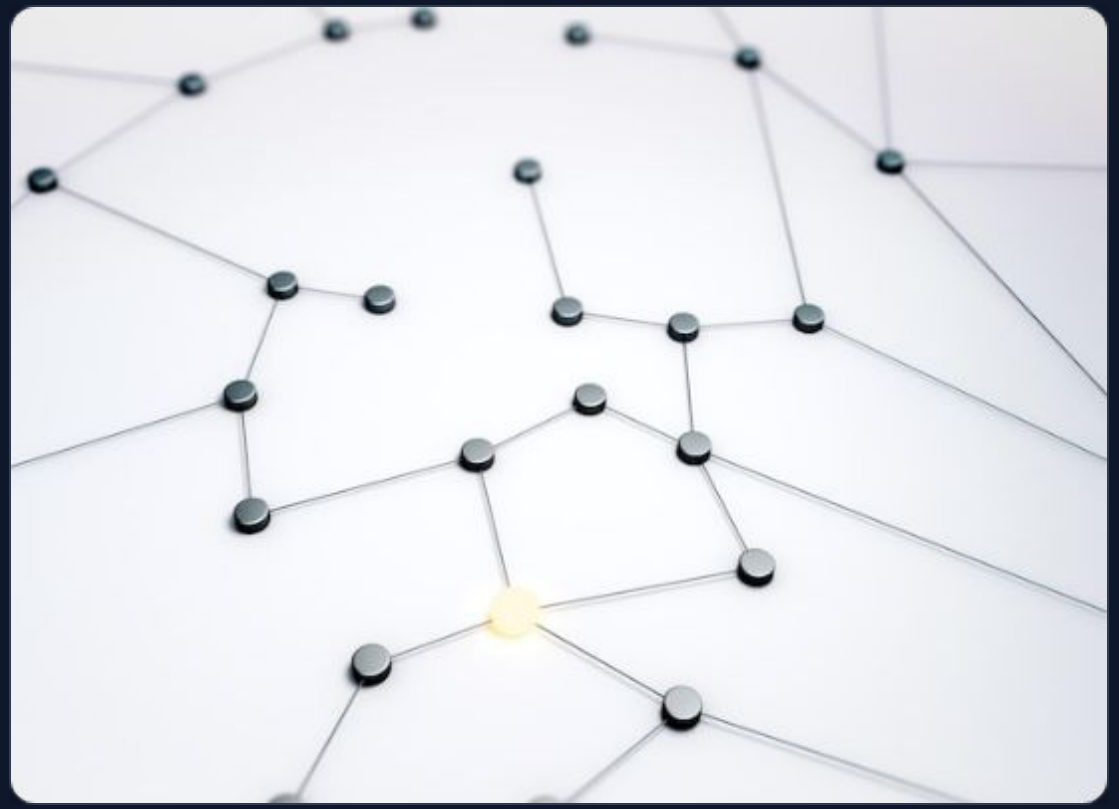
Entity-Relationship (ER) Model Concepts

A comprehensive guide to logical database design and architecture.

Introduction to ER Models

The **Entity-Relationship (ER) model** is a conceptual framework used to represent the data and relationships in a database.

- Helps design databases by defining entities, attributes, and relationships.
- Represents database schema graphically, allowing easy conversion to relational tables.
- Models real-world objects to visualize data structures intuitively.
- Provides a standardized, easy-to-understand approach even for non-technical users.



Components of an ER Diagram



Entities

Objects or things in the real world that are distinguishable from other objects.

Examples: "Student", "Course", "Teacher".



Attributes

Characteristics or properties that describe an entity.

Examples: For a "Student" entity, attributes include Student_ID, Name, and Age.



Relationships

Associations or links established between different entities.

Example: A "Student" **enrolls in** a "Course".

Symbols Used in ER Modeling

- **Rectangles:** Represent Entities (e.g., Student, College).
- **Ellipses / Ovals:** Represent Attributes belonging to entities (e.g., Name, Age).
- **Diamonds:** Represent Relationships among different entities (e.g., StudyIn).
- **Lines:** Act as connectors, linking entities to their attributes or to relationships.

Entity Sets & Entity Types

What is an Entity Set?

It is a collection of similar types of entities that share the same properties or attributes. In simpler terms, it's a group of entities that belong to the same type.

- If "Student" is the entity type, then individuals like "John" or "Emily" are entities.
- The collection of all students (John, Emily, etc.) forms the **Entity Set**.

Strong Entity

Can exist independently and has a primary key that uniquely identifies each instance. Represented by a standard rectangle. (e.g., Employee)

Weak Entity

Cannot exist independently and relies on a strong entity. Does not have its own primary key. Represented by a double rectangle.

Types of Attributes: Value Cardinality

1. Single-Valued Attribute

Definition: An attribute that holds only a single value for a particular entity instance.

Example: Age of a person, or Employee_ID of an employee.

Representation: In ER diagrams, it is represented by a standard single ellipse/oval.

2. Multi-Valued Attribute

Definition: An attribute that can have multiple values for a single entity instance.

Example: Phone_No of a person (since a person may have more than one phone number).

Representation: In ER diagrams, it is represented by a double ellipse.

Types of Attributes: Structural Complexity

3. Simple (Atomic) Attribute

Definition: An attribute that cannot be divided further into smaller, meaningful sub-attributes.

Example: First_Name, Last_Name, Date_of_Birth.

Representation: Represented by a standard single oval.

4. Composite Attribute

Definition: An attribute that can be divided into smaller sub-attributes, each representing a part of the whole.

Example: Address (can be divided into Street, City, State, and Zip_Code).

Representation: Represented by an oval connected to smaller ovals representing its sub-attributes.

Types of Attributes: Identification

5. Key Attribute

Definition: An attribute that uniquely identifies an entity within an entity set. It serves as the primary key.

Example: Roll_No for a student, Employee_ID for an employee.

Representation: Represented by an oval with the attribute name underlined.

6. Non-Key Attribute

Definition: An attribute that does not uniquely identify an entity but provides additional descriptive information.

Example: Name or Address for a student.

Representation: Represented by a single oval without any underline.

Other Attribute Classifications



7. Derived Attribute

An attribute whose value can be calculated or derived from other attributes.

Example: Age (derived from Date_of_Birth).

Notation: Dashed oval.



8. Required Attribute

An attribute that must have a valid value for each entity in the set.

Example: Employee_ID.

Notation: Generally assumed; no specific visual mark.



9. Optional Attribute

An attribute that may or may not have a value (can be NULL).

Example: Middle_Name.

Notation: Handled in documentation or schema rules.